A primer on structured decision making for invasive species (using dynamic programming)

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Invasive species (IAS) pose significant ecological, economic, and public health challenges. Robust methods are needed for guiding effective management. Overall, we hope to demonstrate the flexibility and utility of structured decision making in invasive species management.

Target journal: either Biological invasions (invasion notes?) or NeoBiota.

Add a shiny for each case study so that folks can play around?

Keywords: Invasive species, Structured decision making, Adaptive management, Stochastic dynamic programming

Introduction

Invasive species are a significant global issue, with wide-ranging impacts on ecosystems, economies, and public health (Pyšek et al. 2020, Roy et al. 2024).

Effective management of invasive species requires blabla for guiding control efforts and evaluating the success of eradication or regulation programs (Williams et al. 2002, Thompson et al. 2021).

Pitch: Structured decision-making (SDM) and adaptive management (AM) have been repeatedly recommended for the management of invasive species, yet they remain underused in practice. There are many reasons for this gap, one of which is the technical challenge of formally identifying and quantifying trade-offs among management objectives. Here, we contribute to ongoing efforts by illustrating the use of stochastic dynamic programming (SDP) and some of its variants — including active and passive adaptive management and partially observable Markov decision processes (POMDPs) — across different ecological contexts (a single population, a metapopulation, and two interacting populations).

Several papers have recommended or implemented structured decision-making (SDM) for managing invasive species. It's worth providing a short review of this literature.

A primer on structured decision making

Why and what

Stochastic dynamic programming

Example on computer maintenance to explain SDP on a simple example?

Case studies with coypus

Single population

Take wolf example and turn it into some invasive species example?

Two interacting species

Competition or predator-prey, see SDP-species-interactions/ in my biblio repo.

Metapopulation

Acounting for uncertainty

POMDP

Illustrate POMDP, i have some code extending the computer maintenance example from above.

Adaptive management

On a single species ; check out Conroy's book, there is a nice figure illustrating AM.

Discussion

Main messages

Limitations

Perspectives

Acknowledgments

Data availability statement

Data and code are available at https://github.com/oliviergimenez/SDPaper.

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